```
import java.util.Scanner;

public class hillc{

   static String convert(String text, int[][] key) {
      String result = "";

      for (int i = 0; i < text.length(); i += 2) {
         int a = text.charAt(i) - 'A';
         int b = text.charAt(i + 1) - 'A';

         result += (char) ((key[0][0] * a + key[0][1] * b) % 26 + 'A');
         result += (char) ((key[1][0] * a + key[1][1] * b) % 26 + 'A');
      }

      return result;
   }

   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      int[][] key = {{3, 3}, {2, 5}};
      int[][] inverse = {{15, 17}, {20, 9}};

      System.out.print("Enter the message: ");
      String message = sc.nextLine().toUpperCase().replace(" ", "");
```

```
        if (message.length() % 2 != 0)
            message += "X";

        String encrypted = convert(message, key);
        String decrypted = convert(encrypted, inverse);

        System.out.println("Encrypted: " + encrypted);
        System.out.println("Decrypted: " + decrypted);

        sc.close();
    }
}
```